

BPSC TRE 4.0 – Higher Secondary (Class 11–12)

Computer Science

Syllabus tick-checklist / सिलेबस चेकलिस्ट | ☐ = pad liya / tick it when done

Source note: The BPSC website's syllabus page (bpsc.bihar.gov.in/syllabus) does not list a TRE 4.0 syllabus; it is given in Advertisement 15/2026. The notification PDF I downloaded is image-only (no readable text), so this checklist is built from the topic-wise Higher Secondary syllabus PDF published by Adda247 (a coaching site, *not* BPSC). Both files are in this folder. **Verify against the official notification before relying on it.**

बीपीएससी की साइट पर TRE 4.0 का अलग सिलेबस नहीं मिला। यह चेकलिस्ट Adda247 के सिलेबस PDF पर आधारित है — आधिकारिक विज्ञापन 15/2026 से मिलान ज़रूर करें।

Part	Subject	Questions / Marks
I	Language (qualifying)	30 / 30
II	General Studies	40 / 40
III	Computer Science	80 / 80
	Total	150 / 150

Part III – Computer Science (80 marks) – most important

1. Computer fundamentals

- ☐ Hardware: CPU, memory hierarchy, I/O devices
- ☐ Software types, operating systems basics
- ☐ Data representation, number systems and conversions
- ☐ Boolean logic: gates, laws, De Morgan, K-map, adders, MUX, flip-flops

2. Programming

- ☐ Algorithms and flowcharts
- ☐ Variables, data types, operators
- ☐ Control structures: if, loops, switch
- ☐ Functions and recursion
- ☐ Strings, arrays/lists
- ☐ File handling
- ☐ Languages used in school books: C / C++ / Python

3. Object-oriented programming

- ☐ Classes and objects
- ☐ Encapsulation, abstraction, inheritance, polymorphism
- ☐ Constructors / destructors, basic software design

4. Data structures

- ☐ Stacks (push/pop, postfix evaluation)
- ☐ Queues (circular, deque, priority)
- ☐ Linked lists
- ☐ Trees: BST, traversals, balanced trees, heaps
- ☐ Graphs: BFS, DFS, spanning tree
- ☐ Searching: linear, binary, hashing

- ☐ Sorting: bubble, selection, insertion, merge, quick
- ☐ Time complexity, Big-O

5. Databases

- ☐ Relational model, ER diagram
- ☐ Keys: primary, foreign, candidate, super
- ☐ SQL: DDL, DML, DCL, TCL, joins, group by / having
- ☐ Normalisation: 1NF, 2NF, 3NF, BCNF
- ☐ Transactions, ACID, database applications

6. Computer networks

- ☐ LAN / MAN / WAN, topologies, devices
- ☐ OSI and TCP/IP models
- ☐ Protocols: HTTP, FTP, SMTP, DNS, DHCP, TCP, UDP
- ☐ IP addressing, ports
- ☐ Internet, web, network security

7. Cybersecurity, privacy, ethics, AI, ML, cloud, IoT

- ☐ Malware types, phishing, firewall, encryption, IT Act
- ☐ Digital privacy and ethics, plagiarism, copyright
- ☐ Artificial intelligence and machine learning basics
- ☐ Cloud computing (IaaS, PaaS, SaaS), IoT, big data, blockchain

8. Also asked in earlier papers (not named in the syllabus text, but safe to cover)

- ☐ Operating system: scheduling, deadlock, paging, segmentation, thrashing, file systems
- ☐ Algorithm design: divide and conquer, greedy, dynamic programming
- ☐ MS Office, internet and e-mail basics

Part II – General Studies (40 marks)

Elementary Mathematics

- ☐ Number system, HCF/LCM, simplification
- ☐ Ratio and proportion, percentage
- ☐ Profit, loss, discount, simple and compound interest
- ☐ Average, time and work, time-speed-distance
- ☐ Algebra, geometry, mensuration
- ☐ Data handling: tables, graphs, mean/median/mode, probability

Mental Ability

- ☐ Analogy, classification, series
- ☐ Coding-decoding, blood relations, direction sense
- ☐ Ranking, syllogism, Venn diagrams, calendar, clock, puzzles

General Awareness

- ☐ India and neighbours, important institutions
- ☐ Current affairs, schemes, awards, important days
- ☐ Indian history and culture
- ☐ Indian polity: Constitution, rights, Parliament, judiciary
- ☐ Indian economy: banking, budget, inflation, schemes
- ☐ Sports
- ☐ Environment and ecology

General Science

- ☐ Physics: motion, force, energy, heat, sound, light, electricity
- ☐ Chemistry: matter, atoms, acids/bases/salts, metals, carbon compounds
- ☐ Biology: cell, nutrition, human body, diseases
- ☐ Ecosystem and environment

Social Science, Geography and Indian National Movement

- ☐ History, civics, economics, social issues
- ☐ Solar system, Earth, maps, physical, Indian and human geography
- ☐ 1857 revolt, reform movements, Congress, Swadeshi, Gandhian era, Independence

Tip: Bihar GK and teaching-related questions are commonly asked in GS — add Bihar history, geography, schemes and current affairs to your list.

Part I – Language (30 marks, qualifying)

English (8 questions in the 2024 paper)

- ☐ Reading comprehension
- ☐ Parts of speech, tenses, articles, subject-verb agreement
- ☐ Active/passive, direct/indirect speech
- ☐ Vocabulary: synonyms, antonyms, idioms, one-word substitution
- ☐ Error detection, fill in the blanks, sentence rearrangement

Hindi / Urdu / Bengali (22 questions in the 2024 paper)

- ☐ Gadyansh / padyansh (comprehension)

- ☐ Grammar: sangya, sarvnam, visheshan, kriya, ling, vachan, karak, kaal, vachya
- ☐ Sandhi, samas, upsarg-pratyay
- ☐ Muhavare, lokoktiyan, paryayvachi, vilom, anekarthi
- ☐ Spelling and sentence correction